

# MVP Simulator: MDP Formulation and Initial Results

William Dennis

June 2026

## 1 MDP Formulation

The MVP simulator implements a finite-horizon MDP  $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ :

### 1.1 State Space

$$\mathcal{S} = \{\text{sedentary}, \text{active}\}$$

Two activity states with no additional time-of-day dimension. Time-dependent reward scaling is handled via `reward_multiplier_by_step` (see §1.4 below).

### 1.2 Action Space

$$\mathcal{A} = \{\text{nudge}, \text{idle}\}$$

A binary intervention: send a motivational message (`nudge`) or do not (`idle`).

### 1.3 Transition Function

$P(s' | s, a)$  is a configurable  $2 \times 2 \times 2$  tensor. Rows index current state  $s \in \{\text{sedentary}, \text{active}\}$  and columns index next state  $s'$  in the same order:

$$P_{\text{nudge}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix} \quad P_{\text{idle}} = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$$

Sending a reminder increases  $P(\text{active} | \text{sedentary})$  from 0.1 to 0.3 but decreases  $P(\text{active} | \text{active})$  from 0.6 to 0.5, consistent with Reactance theory (reminders can backfire for already-active users).

Optional `reward_multiplier_by_step`: a per-step scaling factor that can zero out rewards at certain daily steps (soft mask). When absent, all multipliers default to 1.0. The per-step reward is computed as  $\text{base reward} \times \text{multiplier}[t]$  where  $t$  is the step-of-day index.

### 1.4 Reward

The base reward is a vector indexed by next state:

$$r = [0.0, 1.0]^\top \quad (\text{sedentary}, \text{active})$$

The per-step reward at time  $t$  scales by the multiplier:

$$r(s, a, t) = m[t \bmod S] \cdot r(s')$$

where  $S = \text{steps\_per\_day}$ ,  $m \in \mathbb{R}^S$  is `reward_multiplier_by_step` (default  $m_i = 1.0$ ), and  $s'$  is the next state after taking action  $a$  in state  $s$ . Currently  $m_i = 1.0$  for all  $i$ , so  $r(s, a, t) = r(s')$ .

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

## 1.5 Episode Structure

$T = \text{episode\_days} \times \text{steps\_per\_day}$  timesteps. Default:  $T = 90 \text{ days} \times 5 \text{ steps/day} = 450$ .

## 2 Agents

All agents are bandits: they do not condition on state. From the transition matrix,  $E[r \mid \text{nudge}] \approx 0.375$ ,  $E[r \mid \text{idle}] = 0.20$ , and a state-aware policy that nudges only when sedentary would achieve  $\approx 0.429$ . This gap motivates the state-aware agents planned for Phase 2.

### 2.1 Thompson Sampling

Beta-Bernoulli TS with prior  $\text{Beta}(1, 1)$  per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{I}[s'.\text{activity} = \text{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{I}[s'.\text{activity} = \text{sedentary}]$$

Action selected by sampling  $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$  and choosing  $a^* = \arg \max_a \theta_a$ .

### 2.2 Epsilon-Greedy

$\epsilon = 0.1$ . With probability  $\epsilon$ , select uniformly at random; otherwise select  $\arg \max_a Q(a)$ . Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

### 2.3 UCB1

Upper Confidence Bound with exploration parameter  $c = 2.0$ . Selects:

$$a^* = \arg \max_a \left[ Q(a) + c \sqrt{\frac{\ln N}{N_a}} \right]$$

where  $N$  is total steps and  $N_a$  is the number of times action  $a$  was chosen. The confidence bonus shrinks as  $N_a$  grows, naturally shifting from exploration to exploitation. Each action is pulled once before UCB is applied.

### 2.4 Random Baseline

Uniform random action selection. No learning. Serves as lower bound. The mixture chain  $P_{\text{random}} = \frac{1}{2}(P_{\text{nudge}} + P_{\text{idle}})$  gives  $E[r] \approx 0.308$  per step, matching the empirical  $0.301 \pm 0.024$  within one standard deviation.

### 3 Initial Results

All agents run over 10 random seeds (1–10), 450 timesteps each. Results are reported as mean  $\pm$  standard deviation across seeds.

#### 3.1 Standard Config (config/rule\_based.yaml)

No `reward_multiplier_by_step` – uniform 1.0 multiplier on all steps.

| Agent                                          | Total Reward     | Mean Reward/Step  |
|------------------------------------------------|------------------|-------------------|
| Thompson Sampling ( $\alpha_0 = \beta_0 = 1$ ) | $160.3 \pm 10.5$ | $0.356 \pm 0.023$ |
| Epsilon-Greedy ( $\epsilon = 0.1$ )            | $160.8 \pm 15.3$ | $0.357 \pm 0.034$ |
| UCB ( $c = 2.0$ )                              | $148.2 \pm 15.0$ | $0.329 \pm 0.033$ |
| Random                                         | $135.6 \pm 10.7$ | $0.301 \pm 0.024$ |

Table 1: Results on `config/rule_based.yaml` (10 seeds, 450 timesteps). TS and  $\epsilon$ -greedy perform similarly; TS has lower variance. UCB outperforms random but shows higher variance.

#### 3.2 Masked Config (config/rule\_based\_with\_mask.yaml)

`reward_multiplier_by_step`: `[1, 1, 1, 1, 0]` – step 4 (end of day) yields zero reward regardless of outcome. All agents drop proportionally to the masked fraction (1/5 of steps zeroed), confirming the multiplicative reward model.

| Agent                                          | Total Reward     | Mean Reward/Step  |
|------------------------------------------------|------------------|-------------------|
| Thompson Sampling ( $\alpha_0 = \beta_0 = 1$ ) | $126.6 \pm 6.8$  | $0.281 \pm 0.015$ |
| Epsilon-Greedy ( $\epsilon = 0.1$ )            | $127.3 \pm 10.2$ | $0.283 \pm 0.023$ |
| UCB ( $c = 2.0$ )                              | $116.9 \pm 8.6$  | $0.260 \pm 0.019$ |
| Random                                         | $107.8 \pm 6.7$  | $0.240 \pm 0.015$ |

Table 2: Results on `config/rule_based_with_mask.yaml` (10 seeds, 450 timesteps). Reward drops  $\approx 20\%$  across all agents, matching the 1/5 masked fraction. Relative ordering is preserved.

### 4 Known Limitations

- Agents are state-blind bandits, not state-aware RL. State-aware agents are Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.
- No user response model. Burden accumulation and 4 archetypes are Phase 2.
- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.
- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.